

A Synthetic Study on Vector Retrieval Quality

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Abstract

We present a synthetic paper used to test PDF parsing for a Retrieval-Augmented Generation (RAG) system. The paper is laid out in two columns like a typical NeurIPS submission. We include sections, tables, and references to exercise the parser.

1. Introduction

Vector retrieval has become a key building block of modern AI systems.

Common approaches include BM25 keyword search and dense embedding similarity.

Hybrid retrieval combines both signals via reciprocal rank fusion.

2. Method

We split documents into chunks of roughly 500 characters.

Each chunk is embedded with the BGE-small-zh model.

Retrieved chunks are re-ranked by a cross-encoder.

The final top-K chunks are passed to a large language model as context.

3. Experiment

We evaluate on 8 questions across 6 documents.

Hit@1 measures whether the first result is the correct document.

Context precision measures how many of the top-K results are relevant.

Context recall measures whether all the required keywords were retrieved.

4. Results

Hit@1 reaches 1.0 on the simple test set, indicating perfect top-1 ranking.

Context recall also reaches 1.0, meaning no keyword is missed.

Context precision is 0.825 because some retrieved chunks cover adjacent content.

5. Discussion

A two-column layout typically defeats simple text extractors.

Our parser uses x-coordinate clustering to detect columns and reads in proper order.

Tables are preserved as markdown to keep them queryable.

6. Conclusion

We showed that a small set of techniques yields a working RAG pipeline.

Future work could explore parent-child chunking and contextual retrieval.

Table 1. Retrieval Quality Metrics

Model	Hit@1	Precision	Recall
Baseline	0.50	0.40	0.60
+ BM25	0.75	0.60	0.80
+ Rerank	0.90	0.75	0.95
+ Contextual	1.00	0.83	1.00